

# NIRAJ VALU MATERE

Backend Engineer, Python, API, AI Automation, System Design

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## Professional Summary

Software Engineer with over 2 years of experience in end-to-end product development, backend systems, and AI-driven applications. Experienced in designing, building, deploying, and maintaining full-stack product features across frontend, backend, databases, and cloud environments. Proven ability to develop scalable Python services, RAG pipelines and LLM-powered workflows, with a strong focus on product ownership, stakeholder collaboration, and continuous improvement.

## Skills

**Programming:** Python, JavaScript, C++, SQL, Pandas, NumPy.

**Frontend & Backend:** React.js, FastAPI, REST APIs, Authentication, Full-Stack Feature Development.

**Databases & Cloud:** MySQL, PostgreSQL Database Design, AWS, Deployment, Product Operations, and Maintenance.

**Engineering Practices:** System Design, API Design, Caching, Rate Limiting, Performance Optimization, Clean Code, Agile Development.

## Experience

### SMS DataTech

Oct. 2024 – Present

*Software Engineer*

*Tokyo, Japan*

- **Architected a Distributed Monitoring Pipeline:** Re-engineered a single server monitoring system into a horizontally scalable architecture
- **Dynamic Load Management:** Implemented pre-calculation logic and task queuing to estimate required compute load, automatically launching servers to process 4,000 URLs in under 30 minutes, **13x increase** in throughput.
- **Custom Scheduling Engine:** Developed a granular scheduling feature allowing users to **automate data ingestion** at specific times, days, or monthly intervals.
- **Token-Based Billing Integration:** Transitioned the platform from fixed monthly subscriptions to a **pay-per-use** model.
- **Automated Alerting System:** Built a comprehensive **notification engine** to manage user lifecycle events, including token expiration, purchase confirmations, and pipeline failures.
- **Intelligent Summarization:** Integrated LLMs to automatically summarize scraped web content, enabling **rapid data** synthesis.
- **Semantic Data Filtering:** Developed an **AI-powered search** filter allowing users to query URL differences using natural language prompts.

### SMS DataTech

July 2023 – June 2024

*Software Engineering Intern*

*Tokyo, Japan*

- Optimized backend processing by introducing asynchronous execution and reducing redundant database queries, **decreasing monitoring time** for 100 URLs from **1 hour to 18 minutes**.
- Improved website preview functionality by **reducing load time from 50 seconds to under 10 seconds** and implementing text and image change detection.
- Developed an **NLP-based competitor discovery system** that generates search queries, scrapes results, and ranks websites using cosine similarity within 3 minutes.
- **Led a small development team** through design, implementation, and delivery of a production-ready intelligence system.

## Projects

### ProExamAI - AI Powered Test Platform | *Full-Stack Development* | [Website Link](#)

- Designed, developed, deployed, and maintained an end-to-end AI-powered mock test platform with realistic exam simulations, timed test sessions, result generation, and performance analysis.
- Built full-stack product features across frontend, backend, database, and authentication, including test attempt workflows, answer tracking, AI-generated hints, solutions, and complete AI test analysis.
- Owned the complete product lifecycle from architecture and implementation to production operations, bug fixes, UX improvements, and continuous feature iteration based on user needs.

## Education

### Bachelor of Technology - Computer Science and Engineering

CGPA: 9.01

*Indian Institute of Information Technology (IIIT), Nagpur, India*